



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	P.SwethaNagasri
Student Name	Kallu shresta priya
Roll Number	2520030127
Branch	CSE
Course Outcome	Implemented the 0/1 Knapsack Dynamic Programming algorithm to optimize cargo selection and maximize profit under vehicle capacity constraints.

Case Study Topic:

Bangalore–Mumbai Logistics – 0/1 Knapsack for Cargo Loading

Work Done Summary:

Developed a cargo-loading optimization system using the **0/1 Knapsack Algorithm**. Each shipment has a weight and an associated profit. The objective is to maximize total profit without exceeding the truck's capacity of 40 tonnes. Dynamic Programming was used to evaluate all feasible combinations efficiently and determine the optimal set of consignments.

Implementation Details:

- **Language Used:** Java
- **Data Structure Used:** 2D Dynamic Programming Table
- **Algorithm Used:** 0/1 Knapsack
- Each consignment was represented by its weight and profit.
- A DP table of size $(n+1) \times (W+1)$ was constructed.
- For every item and capacity:
 - If item weight exceeds capacity, item cannot be selected.
 - Otherwise choose maximum of:
 - Excluding the item
 - Including the item
- Final solution obtained from $dp[n][W]$.
- Backtracking performed to identify selected consignments.

Result: Screenshots / Output

```
PS C:\Users\kallu shresta priya> java Knapsack
Maximum Profit = 190
PS C:\Users\kallu shresta priya> |
```

The cargo-loading system successfully applied the **0/1 Knapsack Algorithm** to maximize total profit while respecting truck capacity constraints. Dynamic Programming ensured efficient computation and optimal cargo selection for logistics planning.

GitHub Link: <https://github.com/kallushrestapriya-byte/DSA-case-studies>

Student Signature	Faculty Signature